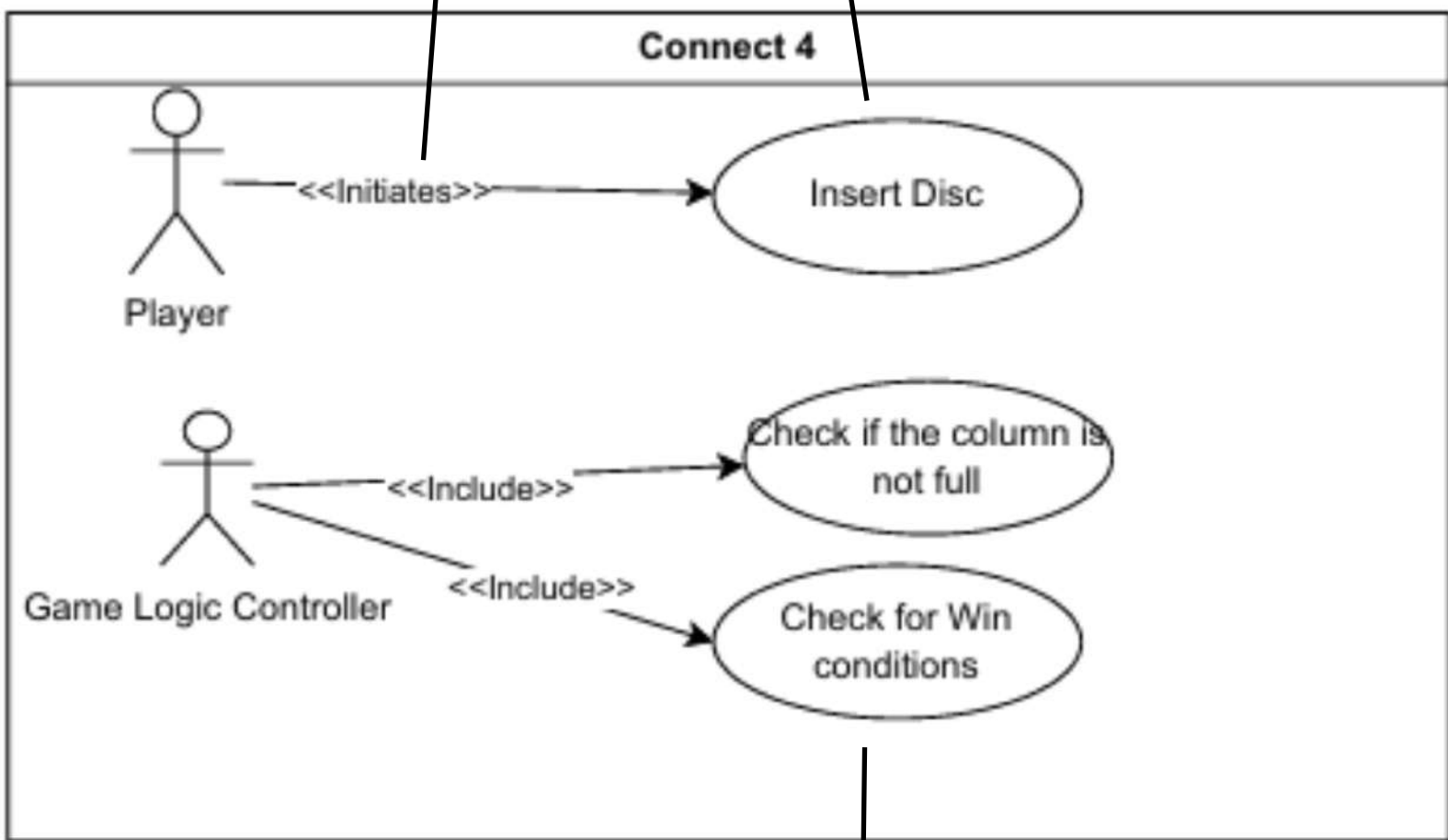


The label "<<initiates>>" is not standard UML

Insert disc and check if column should be connected



Should be check for game over, as a player can win/draw

Connect 4 Use Case Diagram Review Document:

General Critique of the Design:

The diagram presents a core understanding of the interactions involved in the Connect 4 game. However, there are issues with non-standard UML notation and the separation of related processes into distinct use cases that could be more logically implemented.

Diagram Corrections/Changes:

1. Initiates Label:

- **Issue:** The use of "<<initiates>>" is not standard in UML diagrams.
- **Recommendation:** Remove this label and replace it with a standard line to denote association, which simplifies the diagram and adheres to UML standards.

2. Connection of "Insert Disc" and "Check if the column is not full":

- **Issue:** These actions are currently shown as separate but should be connected more logically.
- **Recommendation:** Integrate "Check if the column is not full" directly into the "Insert Disc" process as a precondition.

3. Renaming "Check for Win conditions":

- **Issue:** This use case should encompass checking for all game end conditions, not just winning.
- **Recommendation:** Rename "Check for Win conditions" to "Check for Game Over" to include checks for win, draw, or if the game continues, enhancing clarity.

Feature Request: Restart Game and Bonus Difficulty Mode

- **Restart Game:** At any point during gameplay, if you feel like starting over, this feature allows you and your opponent to agree and restart the game immediately. It's perfect for when you want a clean slate or need to adjust your strategies.
- **Bonus Difficulty with Time Challenge:** This is a timed challenge mode. Each player starts with 20 seconds per move. After each turn, the time limit for the next move decreases by 1 second, stopping at a minimum of 5 seconds. This mode tests your speed and decision-making, making every second count.
- **Expected Impact:** These features increase the dynamics of gameplay. The 'Restart Game' feature offers a fresh start anytime both player needs it,

enhancing flexibility. The 'Bonus Difficulty with Time Challenge' brings an exciting rush, perfect for players looking for a challenge beyond the strategic depth of the game.

- **Implementation/Suggestions:** Implementing these features will involve:
 - Adding a "Restart Game" option accessible during gameplay, requiring mutual consent for activation to maintain fair play.
 - For the timed challenge, implement a countdown timer visible during each player's turn, with automatic decrement settings that adjust the time after every move. Ensure the timer cannot go below 5 seconds to maintain a reasonable response time for all players.

Constructive Feedback:

The use case diagram for Connect 4 could greatly benefit from clearer labeling and improved organization to enhance readability. Currently, the use of labels like "<<initiates>>" diverges from standard UML notation, which could lead to confusion or misinterpretation. Adopting universally recognized UML symbols and labels would not only make the diagram more professional but also easier to understand. Additionally, ensuring that all processes are visually distinct yet logically connected can help clarify the flow of actions and decisions in the game, ultimately making the diagram more intuitive and user-friendly. This approach will help maintain consistency throughout the diagram, reinforcing its utility as a reliable design tool.